Study Schedule

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Mode: Full Pipeline

1. Problem Formulation

Problem Type: Find

Domain: Graph Theory / Combinatorics

Universe

- Assignment: \{Linear Algebra, Abstract Algebra, Probability, Statistics, Real Analysis, Discrete
- Color Map: \{Linear Algebra, Abstract Algebra, Probability, Statistics, Real Analysis, Discrete M
- Graph Stats: \{vertices, edges, max_degree, clique_number\}

Variables

Symbol	Meaning	Type / Range
x_{ij}	1 if item i assigned to slot j, 0 otherwise	\{0, 1\}

Structure

A student has 6 subjects to study across 5 time slots per day (morning, late-morning, afternoon, late-afternoon, evening). Some subjects share prerequisites or overlapping content and should not be studied in adjacent time slots to avoid confusion. Create a weekly study timetable by assigning each subject to a time slot such that conflicting subjects never occupy neighboring slots.

Real-World Mapping

Real-World Concept	Mathematical Object
assignment	x: solution vector (assignment)
total score/cost	objective function value

Constraints

(1)

$$x_{ij} \in \{0, 1\} \quad \forall i, j$$

binary decision variables

Objective

$$\mathrm{optimize}\ f(x)$$

Approach

Suggested approach: Backtracking Distance-2 Graph Coloring

Complexity class:

Available tools: Backtracking Distance-2 Graph Coloring

2. Solution

Answer:	6-element assignment: Linear Algebra -> Morning (8-10), Abstract Algebra
Objective Value:	3
Optimal:	Yes
Feasible:	Yes
Algorithm:	Backtracking Distance-2 Graph Coloring
Time:	0.0000s
Certificate:	Solver reports optimal.

Solution Details

Assignment:
  Linear Algebra -> Morning (8-10)
  Abstract Algebra -> Afternoon (1-3)
  Probability -> Morning (8-10)
  Statistics -> Afternoon (1-3)
  Real Analysis -> Afternoon (1-3)
  Discrete Math -> Morning (8-10)

Objective value: z* = 3

Method: Graph coloring on a conflict graph. Vertices represent subjects, edges represent conflicts (subjects that should not be in adjacent slots). Colors represent time slots. Uses NetworkX's greedy_color with the largest-first strategy. An adjacency constraint is enforced by expanding the conflict graph: if subjects A and B conflict, we add edges not just between A-B but also between any subjects whose assigned slots would be adjacent.

Verification

✓	Feasibility	All constraints satisfied
✓	Optimality	Backtracking Distance-2 Graph Coloring

3. Interpretation

The Question

A student has 6 subjects to study across 5 time slots per day (morning, late-morning, afternoon, late-afternoon, evening).

The Answer

The optimal solution has objective value z* = 3.

What This Means

- All 6 subjects assigned to time slots (colors 0-4)
- No two conflicting subjects share adjacent time slots
- A printable weekly timetable grid

Graph coloring on a conflict graph. Vertices represent subjects, edges represent conflicts (subjects that should not be in adjacent slots). Colors represent time slots. Uses NetworkX's greedy_color with the largest-first strategy. An adjacency constraint is enforced by expanding the conflict graph: if subjects A and B conflict, we add edges not just between A-B but also between any subjects whose assigned slots would be adjacent.

Recommendations

1. The solution is provably optimal; no further improvement is possible within this model.

Limitations

- Model assumes deterministic parameters; real-world variability not captured.
- Solutio